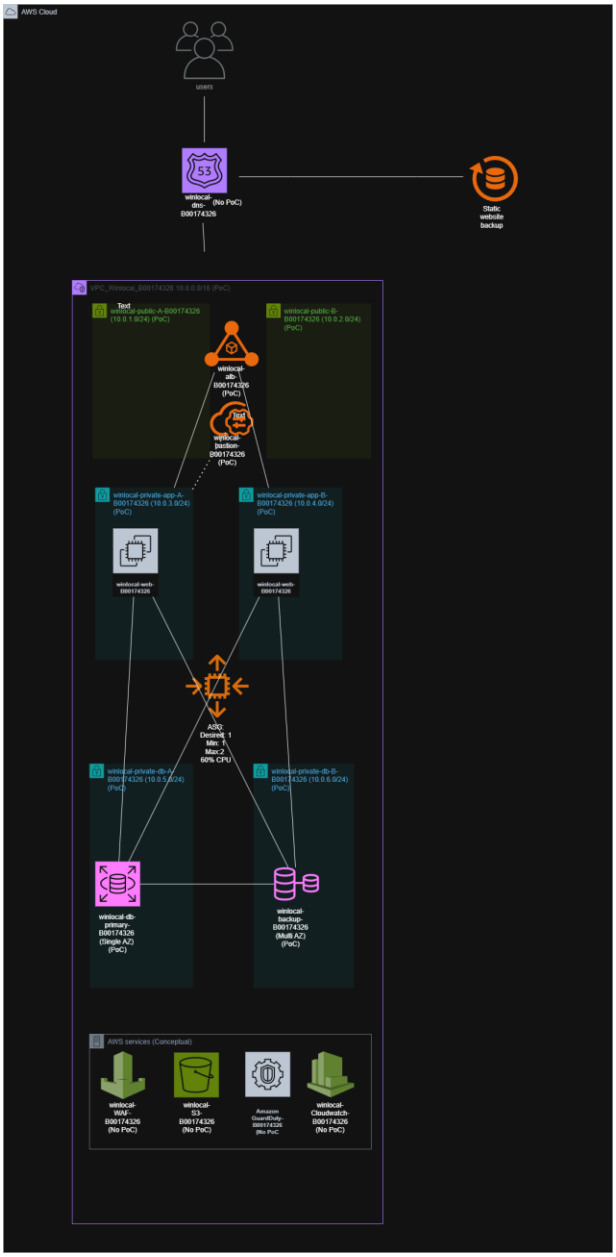
Task 1

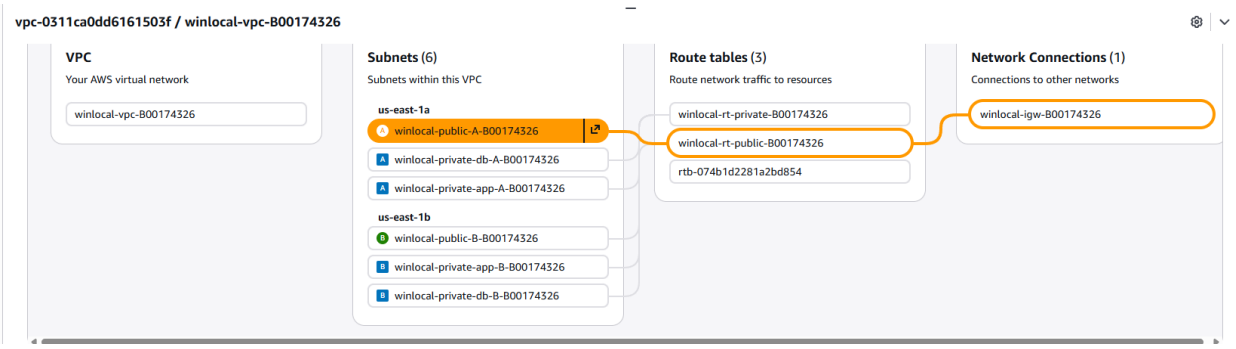


The diagram above shows my AWS architecture for WinLocal Giveaways. It includes a VPC with public and private subnets across two Availability Zones. The public subnets contain the bastion host and Application Load Balancer. The private app subnets contain EC2 web servers behind an Auto Scaling group. The private DB subnets contain an RDS MySQL database (Single-AZ for the PoC, with Multi-AZ shown as conceptual). S3 provides a static backup website for failover. Route 53, AWS WAF, and GuardDuty are marked as "conceptual only" as they were not implemented in the PoC.

Task 2

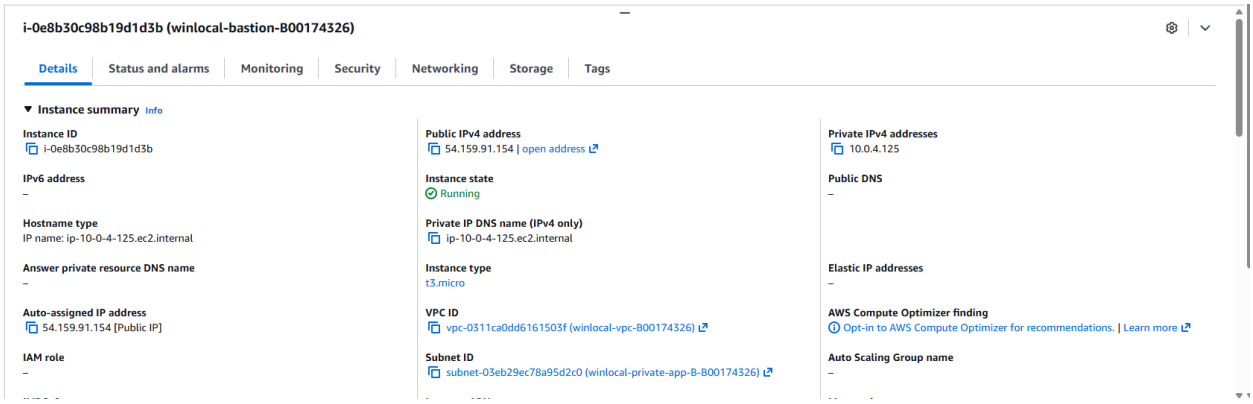
Proof of Concept implementation

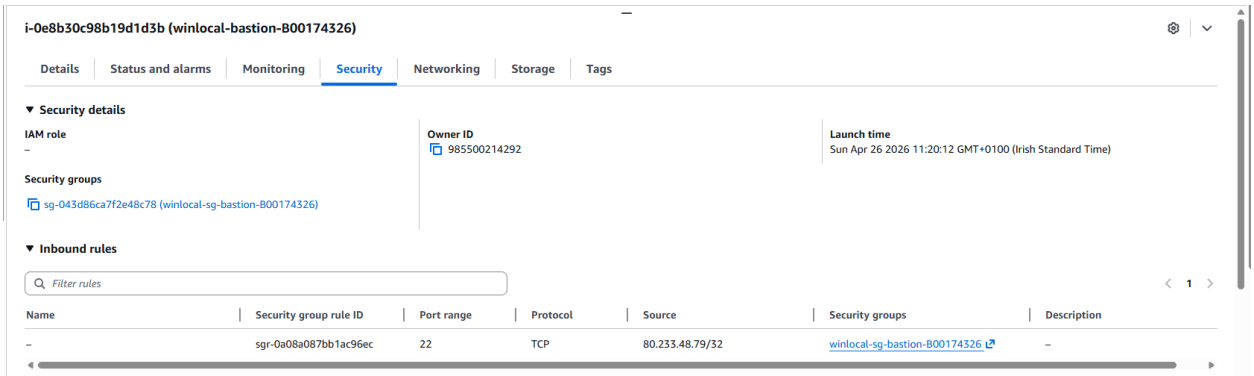
VPC Resource Map



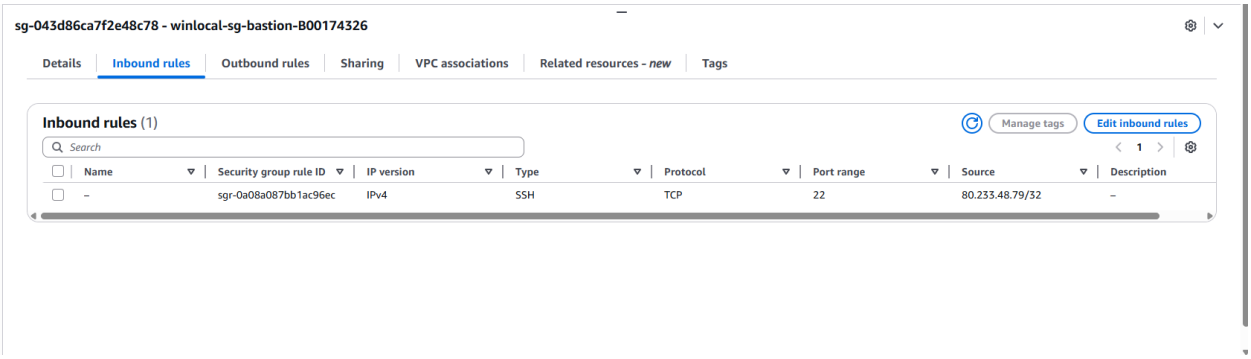
The Figure above shows my VPC (winlocal-vpc-B00174326) with all 6 subnets. The public subnets (10.0.1.0/24 and 10.0.2.0/24) are for the bastion host and load balancer. The private app subnets (10.0.3.0/24 and 10.0.4.0/24) are for the web servers. The Private DB subnets (10.0.5.0/24 and 10.0.6.0/24) are for the RDS database. The internet gateway is attached to allow public traffic in.

Bastion Host



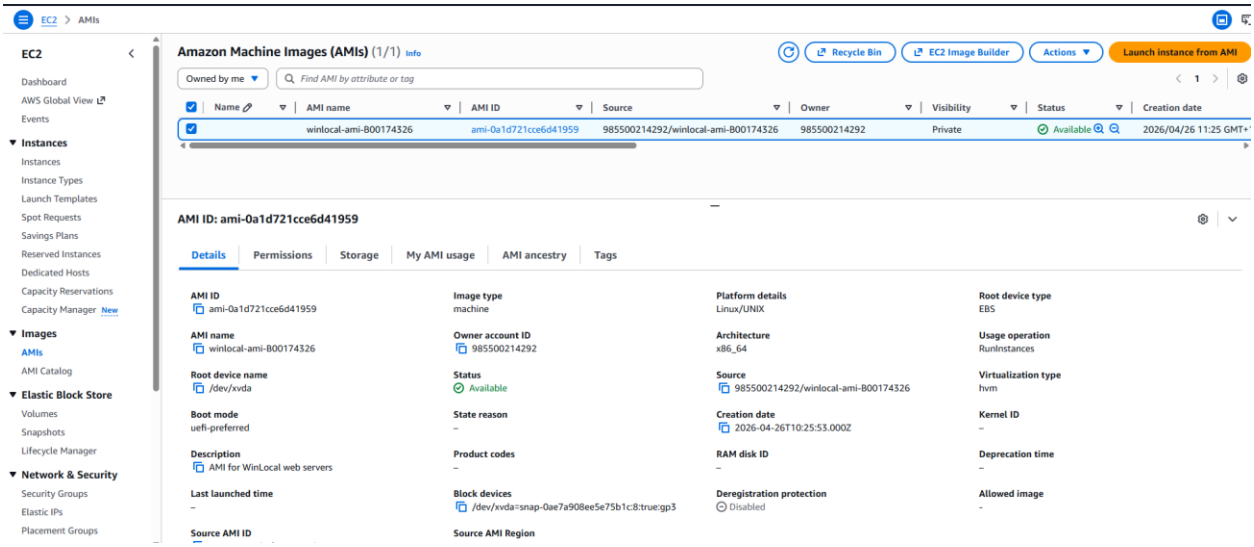


The figures above show my bastion host running with public ip 54.159.91.154. It sits in a public subnet so i can SSH into it from the internet.



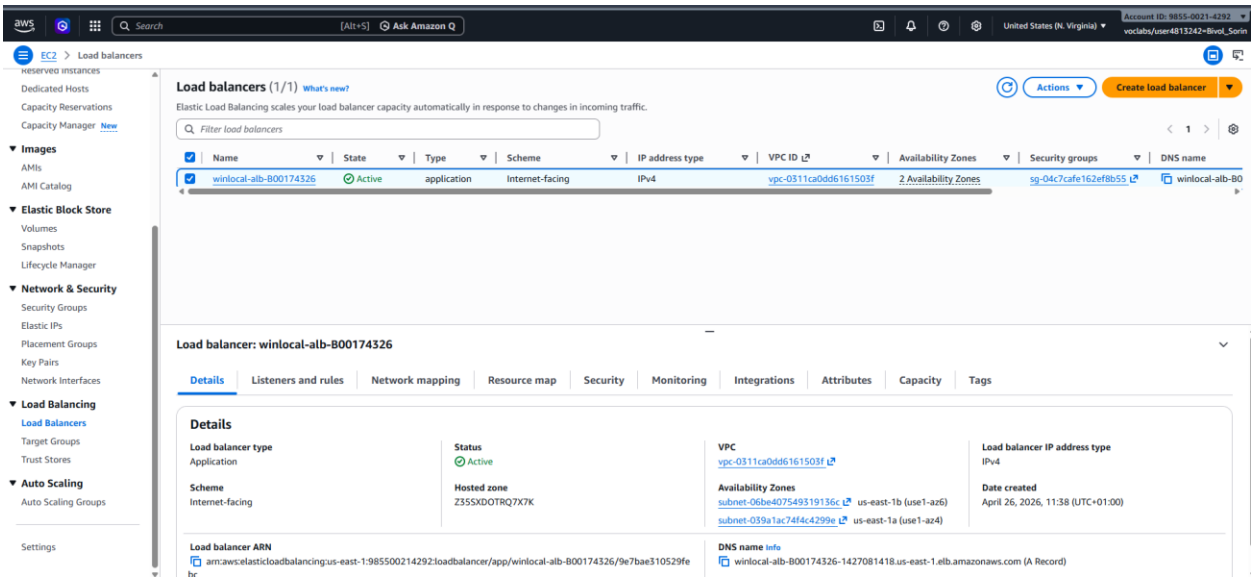
The bastion security group only allows SSH (port 22) from my specific IP address 80.233.48.79/32. Nobody else can try to SSH into it.

AMI Creation



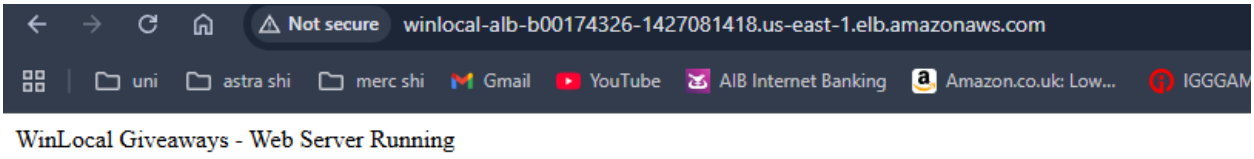
I created a custom AMI called winlocal-ami-B00174326 from a test EC2 instance after installing Apache and PHP. The status is "Available", meaning it's ready to use in my launch template

Load Balancer



My load balancer is active and internet-facing. The DNS name (winlocal-alb-B00174326-1427081418.us-east-1.elb.amazonaws.com) is what users type to access the website

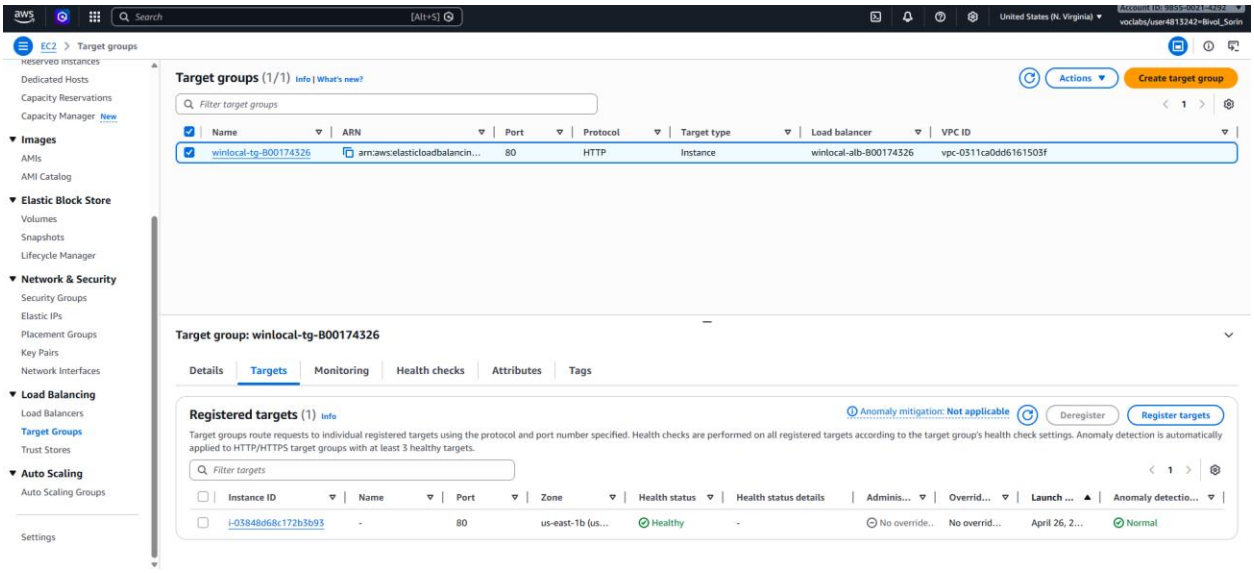
Web Server Running



WinLocal Giveaways - Web Server Running

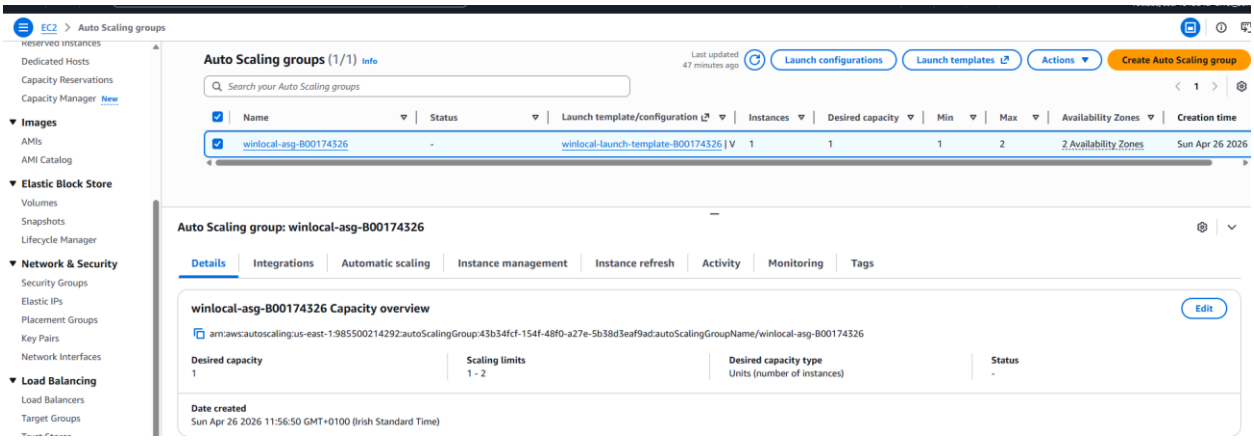
When I paste the ALB DNS name into a browser, I see "WinLocal Giveaways - Web Server Running". This proves the load balancer is sending traffic to my EC2 instance and the web server is working.

Target Group Health



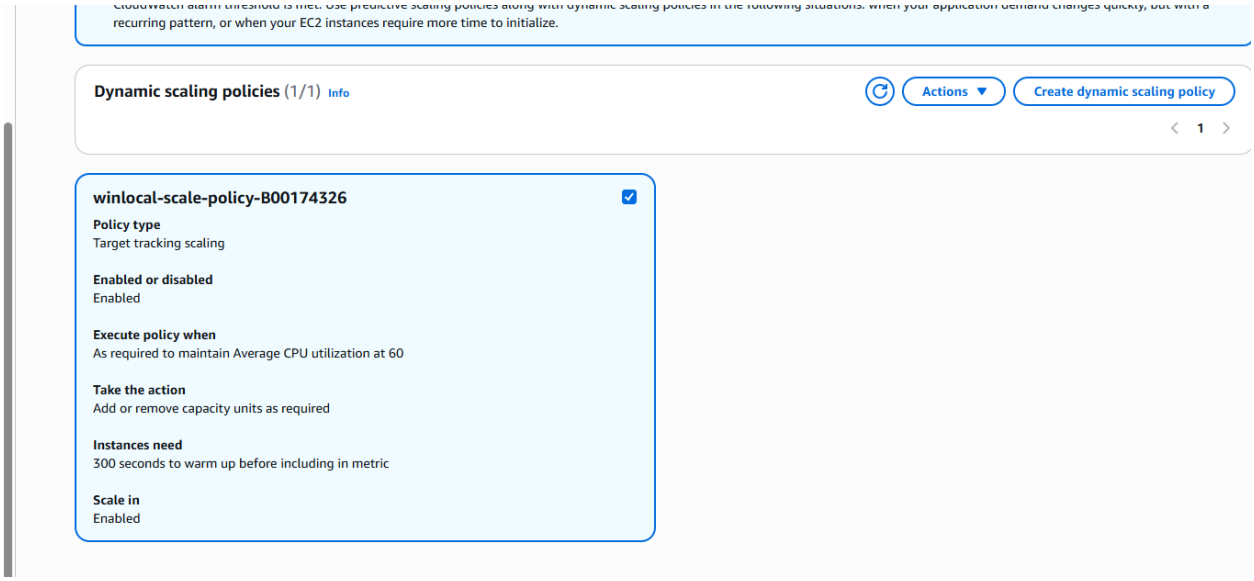
This screenshot shows my target group with one registered EC2 instance. The health status is "Healthy", meaning the load balancer will send traffic to it.

Auto Scaling Group



My Auto Scaling group has desired capacity 1, minimum 1, and maximum 2. This means it always keeps at least one web server running and can scale up to two when needed.

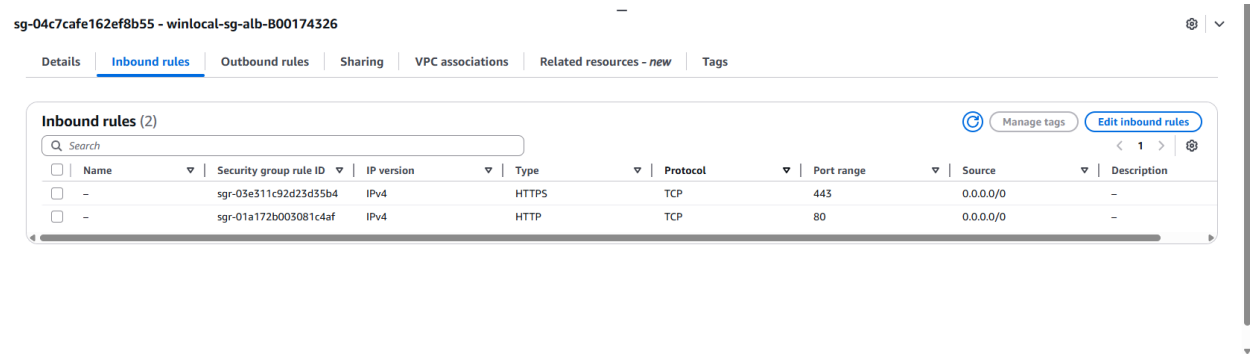
Dynamic Scaling Policy



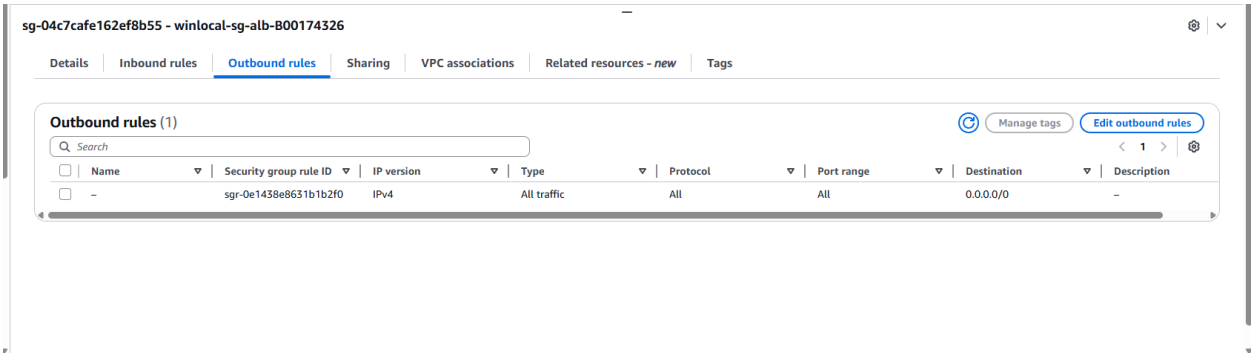
This shows my scaling policy. When average CPU usage hits 60%, Auto Scaling automatically adds a second instance. When usage drops, it scales back down. The policy is enabled and working.

Security Groups

ALB security group

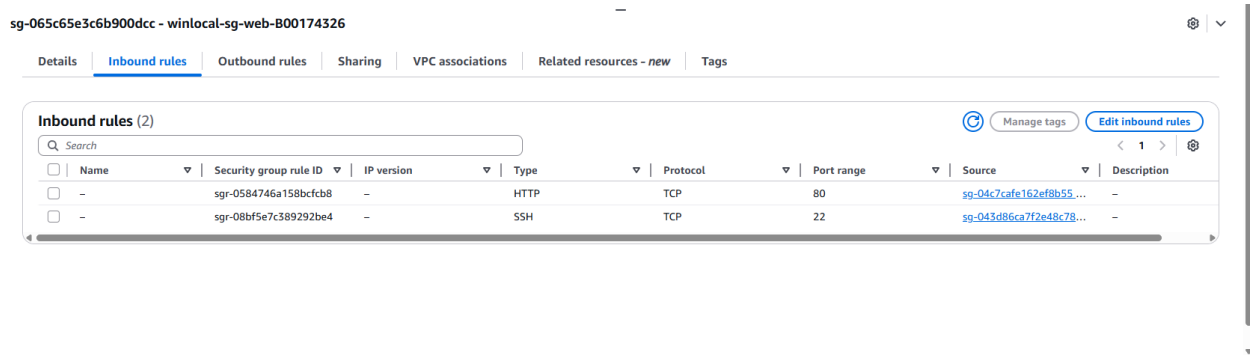


The ALB security group allows HTTP (port 80) and HTTPS (port 443) from anywhere (0.0.0.0/0). This lets users access the website.



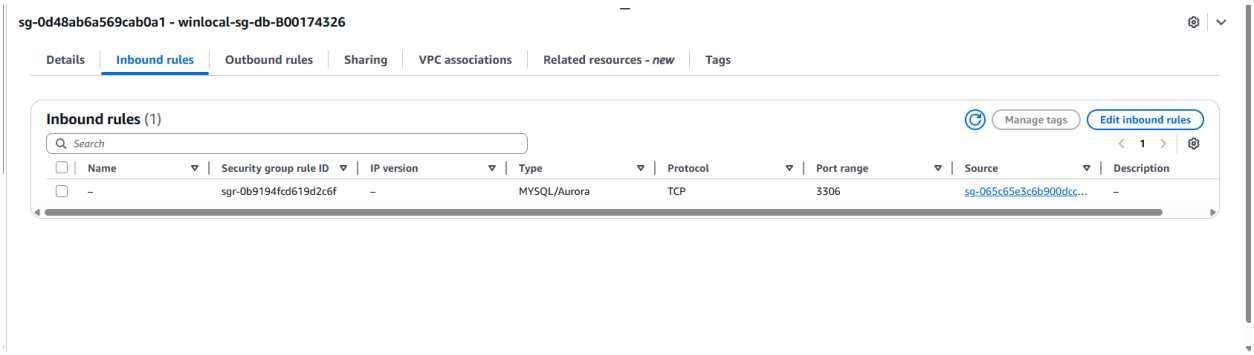
Outbound traffic is allowed anywhere, so the ALB can talk to the web servers.

Web Server Security Group



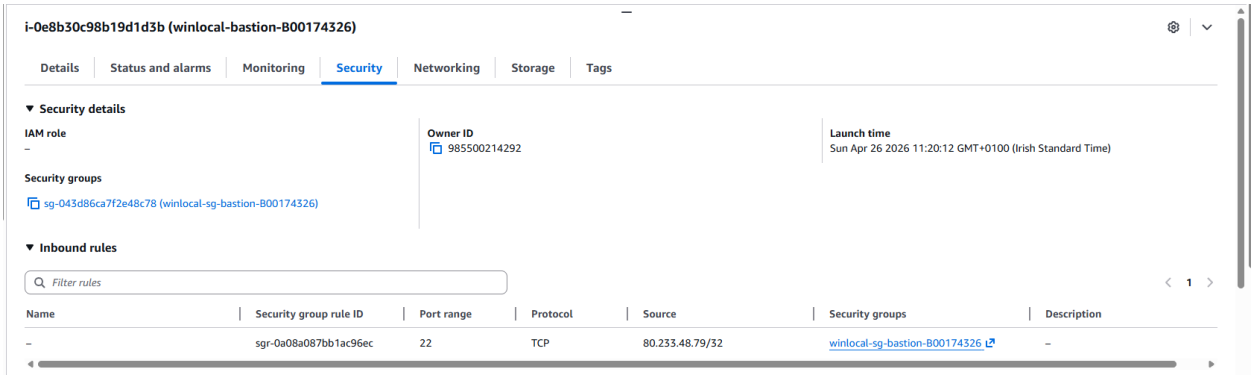
The web security group allows HTTP traffic only form the ALB security Group (not from the internet directly). It also allows SSH from the bastion security group so i can log in through the bastion host.

Database Security Group



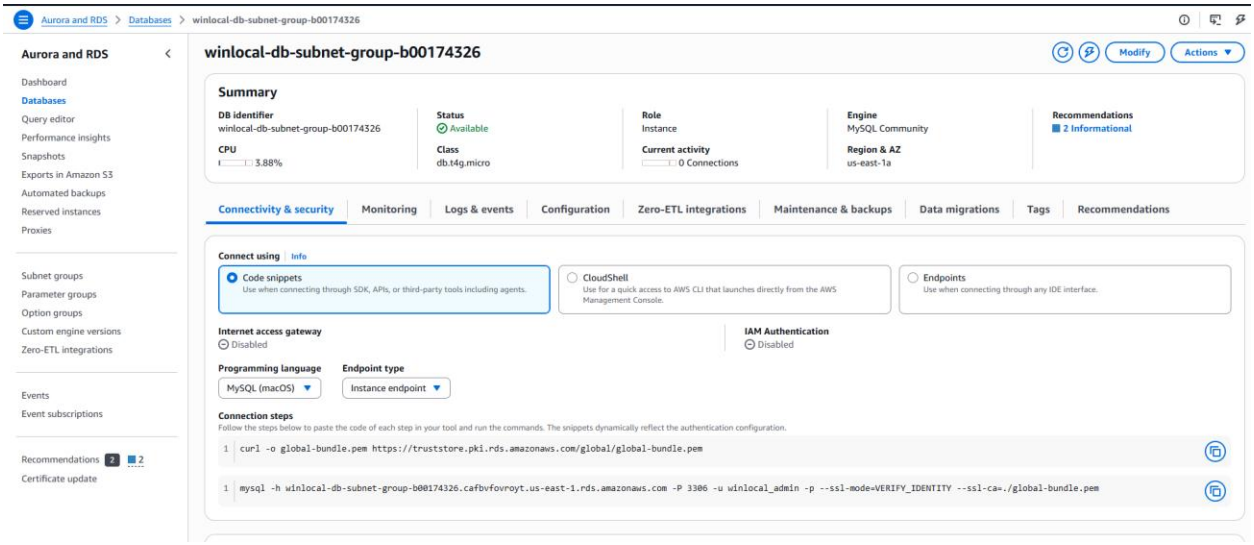
The database security group allows MySQL (port 3306) only from the web security group. This means that only my web servers can connect ot the database – nobody else.

Bastion Host Security Group



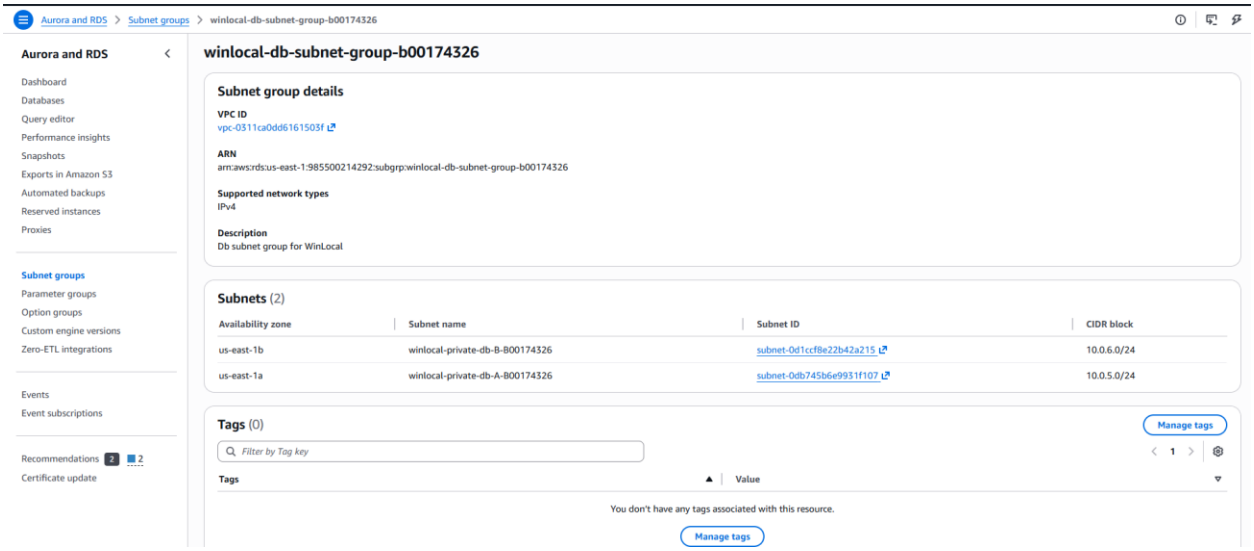
This confirms that the bastion host is using the correct security group (winlocal-sg-bastion-B00174326) with SSH allowed only from my IP.

RDS Database



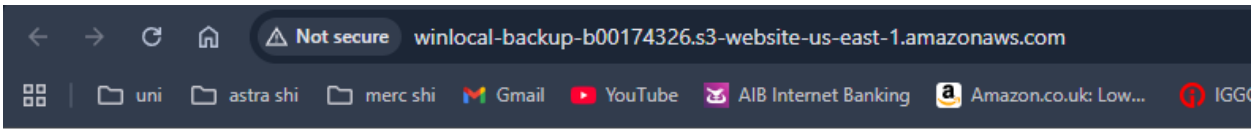
My RDS database is MySQL running on db.t4g.micro. The status is “Available”. Internet access gateway is disabled (Not publicly accessible) so only resources inside VPC can connect to it.

RDS Subnet Group



The DB subnet group contains two private DB subnets (10.0.5.0/24) and (10.0.6.0/24) across both availability zones. This is where the database lives, in a private subnet with no direct internet access.

Static Backup Website.



WinLocal Giveaways

Competition Unavailable

Our website is currently undergoing maintenance. Please check back soon!

Thank you for your patience.

This is my backup Static website hosted on S3. If the main application ever goes down, users would see "Competition Unavailable" page instead of an error message. The brief asked for a failover mechanism, and this S3 bucket is ready to be used with Route 53 (Conceptual).

Task 3

Service 1: AWS WAF

Winlocal got hacked because atackers got in through the website. AWS WAF would sit in front of my load balancer and filter out bad traffic before it hits the web server.

AWS WAF stops things like SQL injection (where attackers try to mess with the database) and cross-site scripting. Basically, it blocks all kinds of attacks that happened to WinLocal.

I didnt include WAF in my PoC because i only had a budget of 50 dollars and needed to prove the core stuff worked first (VPC, EC2, RDS). WAF costs extra money, so it would be something WinLocal adds later.

Service 2: Amazon GuardDuty

The WinLocal brief states that the attackrs were inside their network for 10 weeks before anyone noticed. That is way too much time. GuardDuty is a monitoring service that watches for suspicious activity.

It looks at things like weird API calls, someone trying to access stuff they shouldn't, or instances talking to known bad websites. It would have caught the ransomware attack earlier.

I didnt build this in my PoC because i was focused on getting the website and database working first. GuardDuty is a security add-on that you enable after the main infrastructure is up.

Task 4

My PoC is pretty efficient. I set up Auto Scaling with 1 instance normally and max 2 instances when traffic spikes. That means I'm not running a bunch of empty servers for no reason.

I used t4g.micro instances which are small and cheap. I also didn't use a NAT Gateway (they cost money and waste computing power)- i just used the bastion host.

For the database, i stopped it when i wasn't working on it so it wasn't draining my power or budget.

If Winlocal wanted to be more sustainable in the future they can:

- Use S3 lifecycle rules to automatically move old data to cheaper storage
- Try Graviton instances (better performances for less power)
- Set their dev environment to auto shut down at night.